Why the Power Spectral Density (PSD) Is the Gold Standard of Vibration Analysis

ANALYSIS 14 COMMENTS

BY STEVE HANLY

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You can't get far in vibration testing and analysis without running across and benefiting from a power spectral density (PSD). PSDs, as the name implies, are used to analyze the vibration environment in the frequency spectrum, but FFTs do that, too. So what is it about PSDs that make them so ideal for vibration?

After reading this article I want you to not only understand what a PSD is, but know how and why you can use it to improve your vibration testing & analysis. Throughout the post I'll use data that can be downloaded here and our free software tool, VibrationData Toolbox, to generate the plots. I'll cover:

- What is a PSD?

By the end of the article you will be able to understand how to effectively utilize PSDs to quantify a vibration environment and have the tools to go out and calculate one for your own vibration testing!

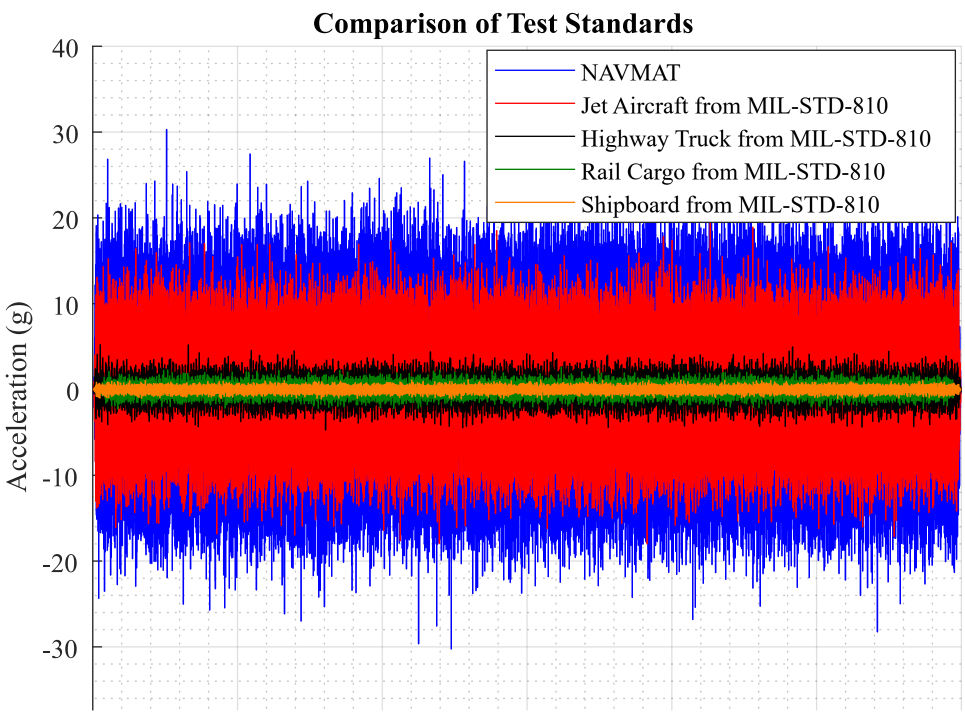
Download the Free VibrationData Toolbox here.

What is a Power Spectral Density (PSD)?

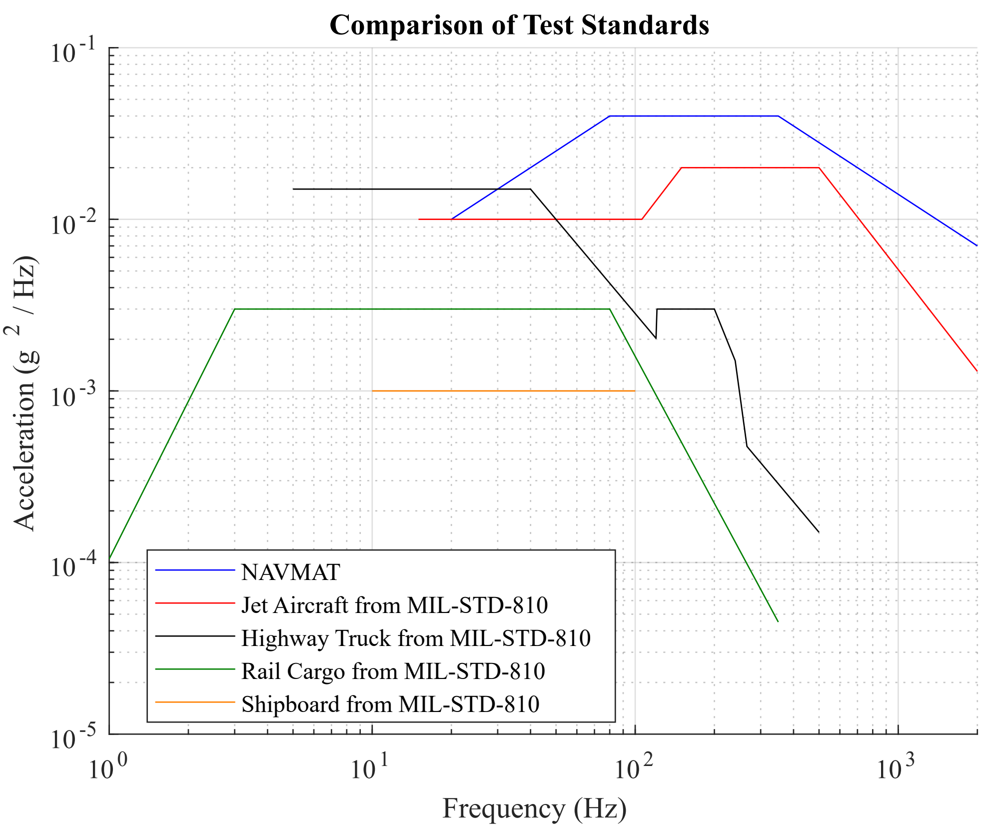
Vibration in the real world is often "random" with many different frequency components. Power spectral densities (PSD or, as they are often called, acceleration spectral densities or ASD for vibration) are used to quantify and compare different vibration environments. Engineers will see PSDs used in test standards such as MIL-STD-810 and others that provide guidance on how to qualify new products & systems for various operational and transportation environments.

I don't want to belabor the math that goes into a PSD (we have our handbook for that!). To illustrate what a PSD is, let's dive into some data! Here are vibration profiles in the time domain over 1 minute for a few different environments:

- NAVMAT (temperature and random vibration screening for Navy contractors)
- Jet Aircraft from MIL-STD-810 (another test standard for military applications)
- Highway Truck from MIL-STD-810
- Rail Cargo from MIL-STD-810
- Shipboard from MIL-STD-810



Looking at this data in the time domain can tell us that the vibration in a jet is likely more severe than rail. But that's only determined from looking at the peak amplitude which isn't always a good indicator. And it doesn't tell us where that energy lies... or what the power *density* is - that's why we have the power spectral density!



Here you can see that the PSD plots power (*g^2 / Hz)* in the y axis as a function of frequency (*Hz)* in the x axis. In this format, we can clearly see the distinction between the vibration environments. So what is this *g^2 / Hz* thing?

To answer that question, let's define how a PSD is calculated. The power spectral density function *XPSD(f)* is calculated from the discrete Fourier transform *X(f)* as

$$X_{PSD}(f) = \lim_{\Delta f \to 0} \left[\frac{1}{2} \frac{X(f)X^*(f)}{\Delta f} \right]$$

The one-half factor is needed to convert the amplitude from $peak^2$/Hz to rms^2/Hz which highlights another benefit of PSDs we'll explore later on.

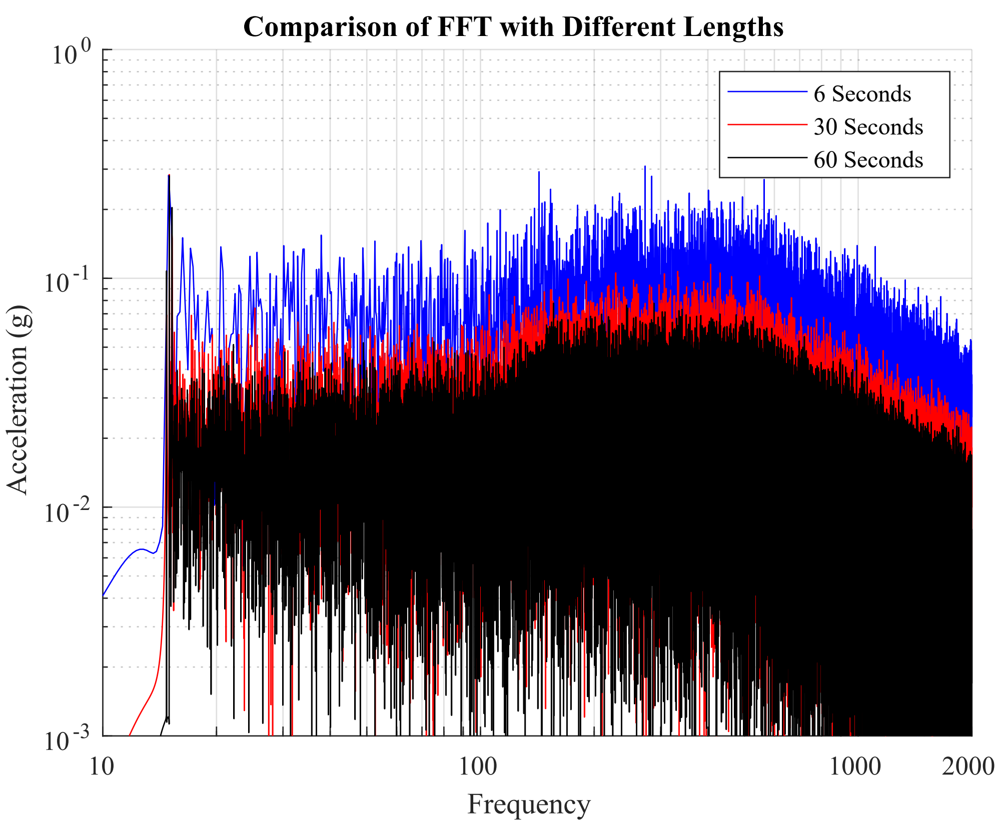
The frequency step is finite in practice and is the inverse of the total measured duration.

PSD is, visit our learn article with a handbook download available.

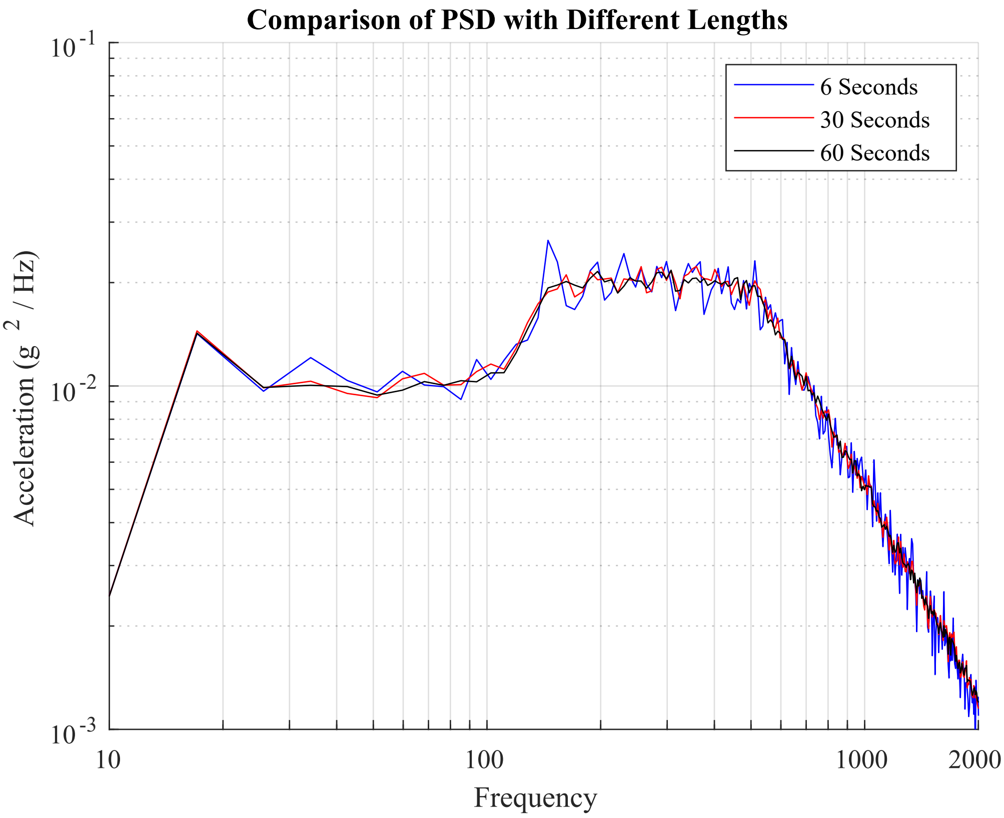
Difference Between PSD and FFT

Although engineers are tempted to use FFTs (fast Fourier transforms) for spectrum analysis, they should really be using (PSDs) power spectral densities. The reason is that PSDs are normalized to the frequency bin width preventing the duration of the data set (and corresponding frequency step) from changing the amplitude of the result. FFTs don't do this!

Let's take that same data from the jet standard and calculate and plot FFTs for different time lengths. Check out how the longer the time series is (and the finer the frequency bin width/step is), the lower the amplitude of the FFT becomes! This means you can't really use FFTs to compare and quantify vibration environments.

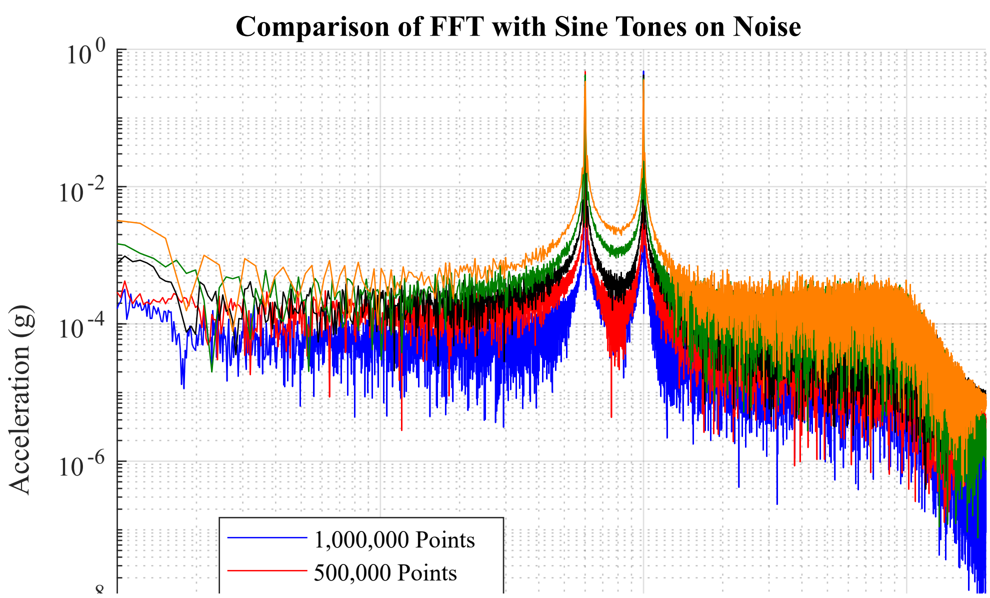


Now let's do that same calculation for the PSD. In this plot I've kept the frequency bin width the same at 8 Hz. These different time lengths do nothing to the amplitude of the resulting PSD because they are normalized to the bin width.

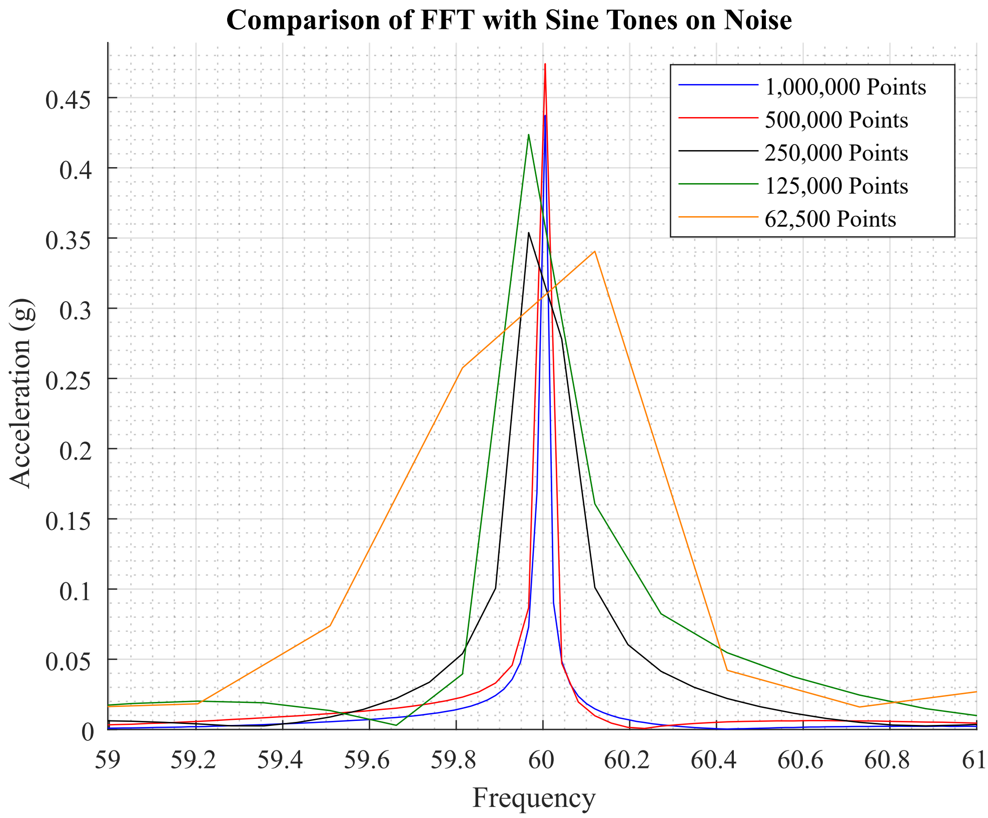


The previous plots, and rest of the article will focus on typical vibration environments that have a broad range of frequency components. But there are some applications where there are very specific frequencies contributing to the signal such as when looking at machinery with rotating equipment. Let's construct a hypothetical waveform of broadband noise with a 0.5g 60 Hz and 100 Hz sine tone on top. This waveform is sampled at 10 kHz for 100 seconds.

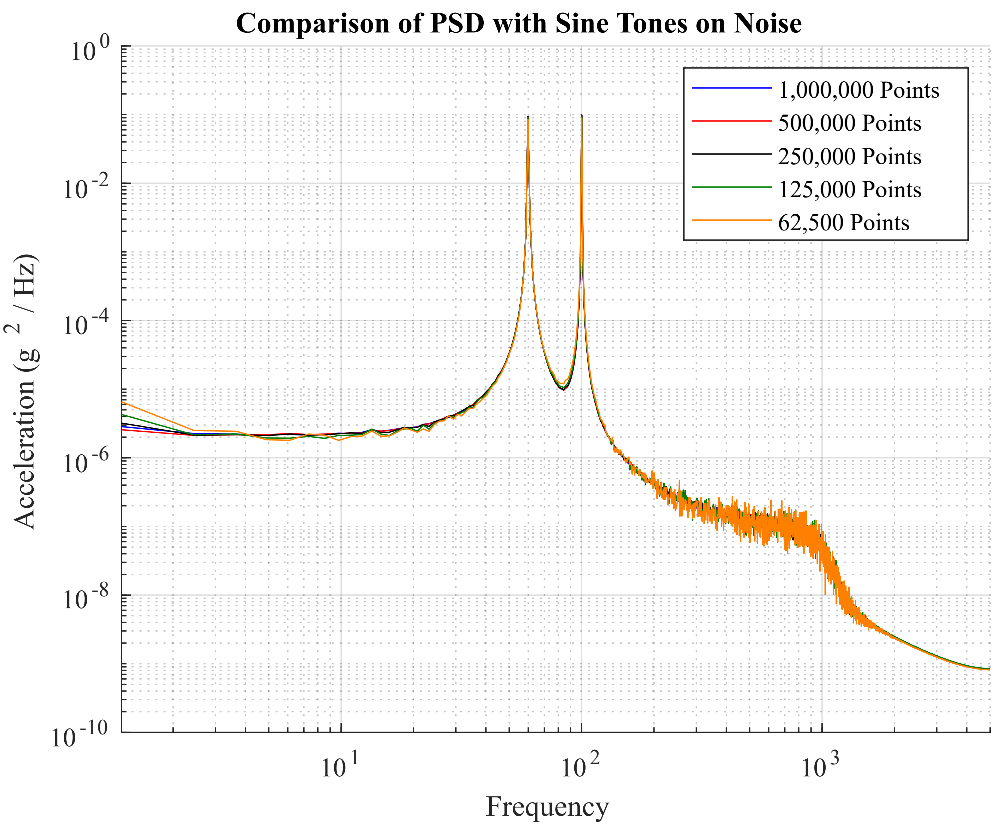
Calculating a FFT at varying lengths shows significant change in the noise floor, but admittedly the amplitude at the peaks seems consistent.

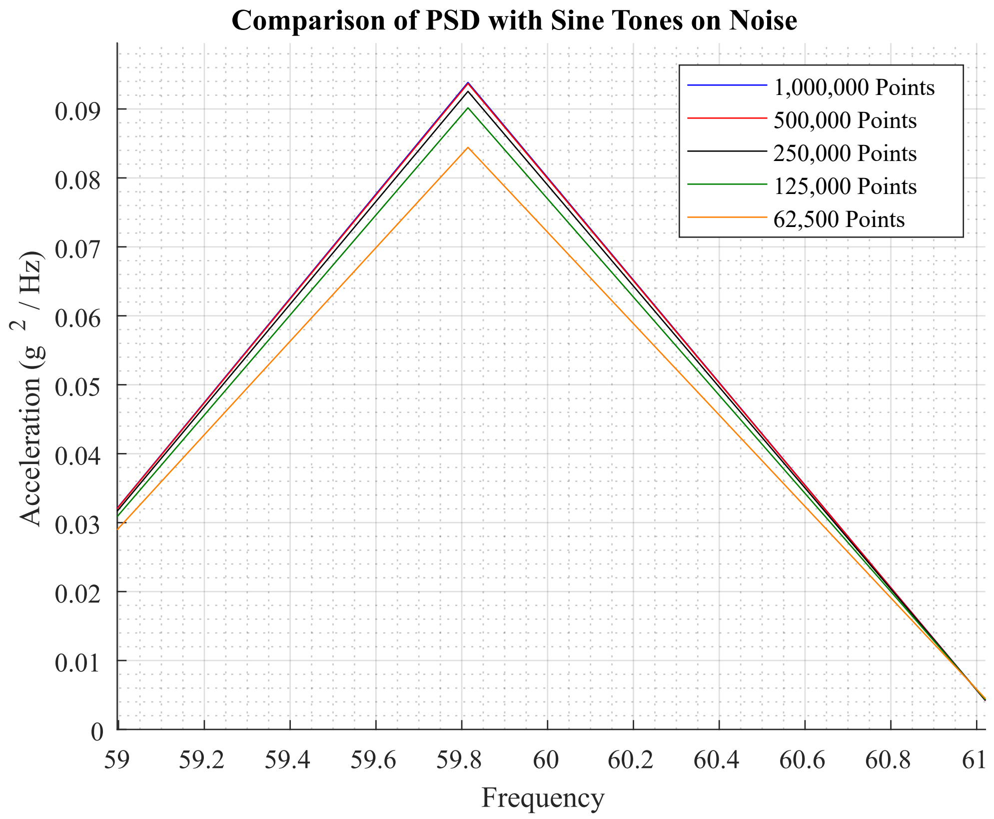


But zooming in on that we can see it is still a little erratic depending on where that bin falls in relation to the pure 60 Hz sine.



Meanwhile the PSD between the different lengths looks identical.

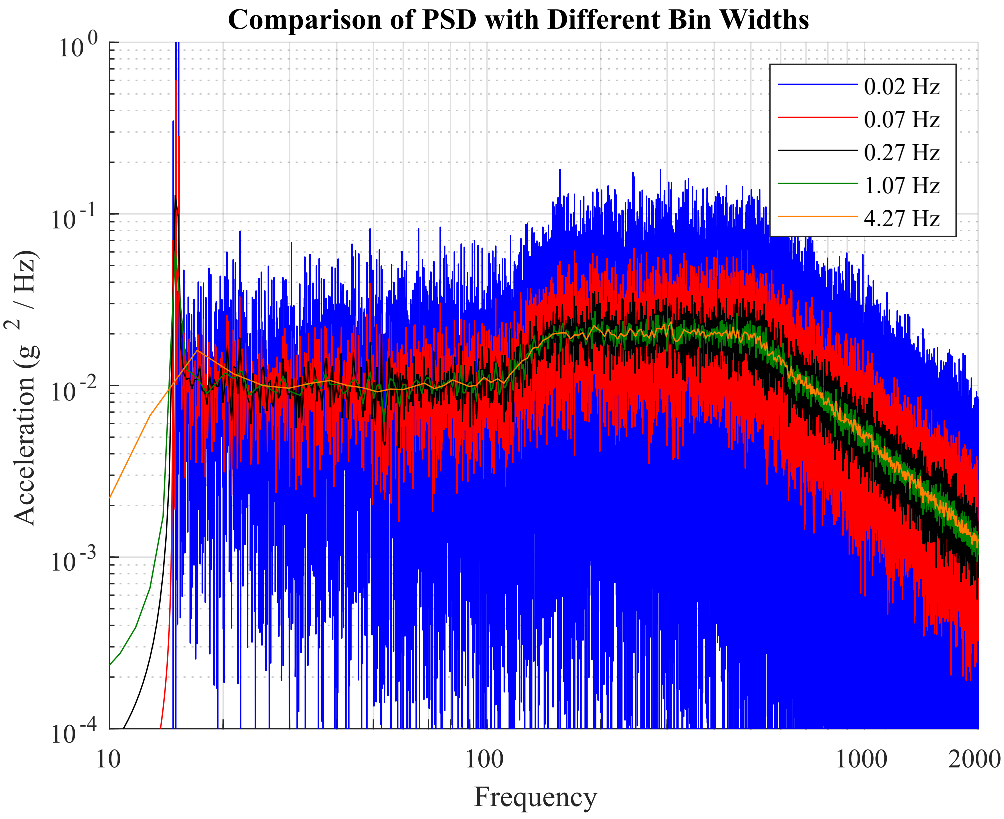




Benefits and Features of PSDs

Adjustable Bin Width to Smooth

You probably noticed in the last section how amazingly clean those PSD were! This is another major benefit of PSDs because they are dependent on that frequency bin width. You can manipulate this to smooth a PSD. The VibrationData Toolbox has a great feature that I highlight in our article on generating a PSD test standard. Here I highlight how changing that bin width from a very fine 0.02 Hz to 4 Hz will drastically smooth the PSD while keeping the overall energy in each frequency accurate and consistent across the whole range.



Download the VibrationData Toolbox here.

Root of Area under a PSD Equates to gRMS

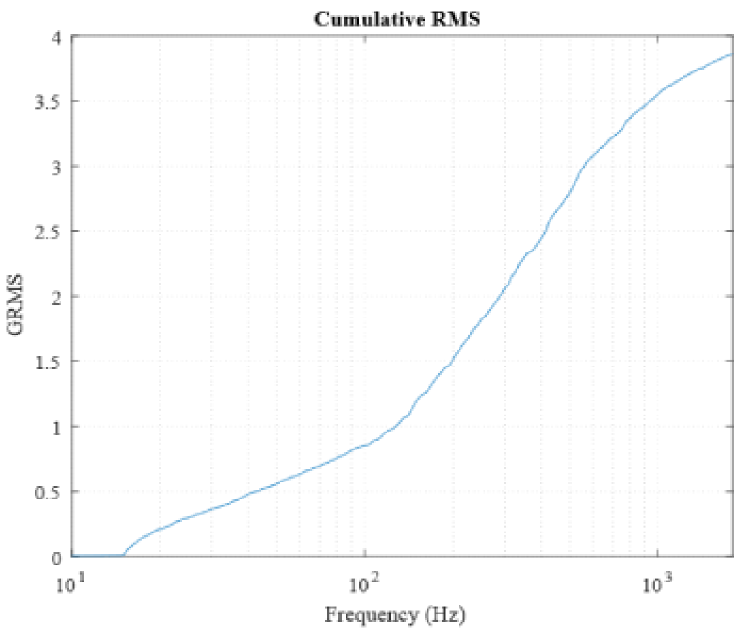
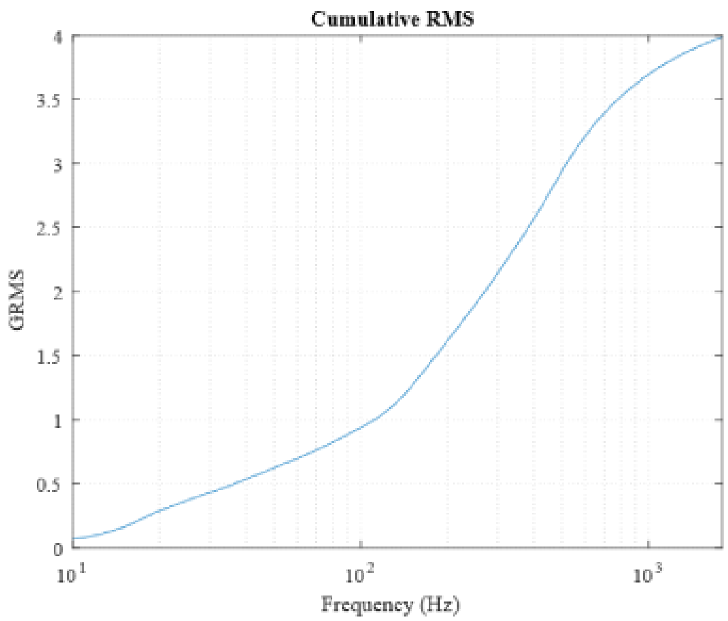
Taking the square root of the area under a PSD will yield the RMS vibration level. To calculate this area requires special integration formulas due to the log-log format of PSD plots. First you need to calculate the slope *N* between each segment:

$$N = \frac{\log[y_2/y_1]}{\log[f_2/f_1]}$$

The area a_i for a segment is then:

$$a_i = \begin{cases} \left[\dfrac{y_i}{f_i^{\,N}}\right]\left[\dfrac{1}{N+1}\right]\left[f_{i+1}^{\,N+1} - f_i^{\,N+1}\right], \text{ for } N \neq -1 \\ \\ \left[y_i f_i\right]\left[\ln\left(\dfrac{f_{i+1}}{f_i}\right)\right], \text{ for } N = -1 \end{cases}$$

Using the previous set of PSDs on the jet cargo vibration environment we can compare the cumulative RMS for the different bin widths of 0.02 Hz and 4.3 Hz. This calculation of the RMS level from the PSD is conveniently done in the VibrationData Toolbox. Although the PSD with the smaller bin width is far nosier, when calculating the gRMS from the respective PSDs (root of area under curve) the two track each other nicely.

Cumulative RMS for 0.02 Hz Bin Width	Cumulative RMS for 4.3 Hz Bin Width
	

Calculating this cumulative RMS is also a helpful way of seeing which frequency components are contributing most of the acceleration amplitude.

Integrating an Acceleration PSD to Velocity & Displacement

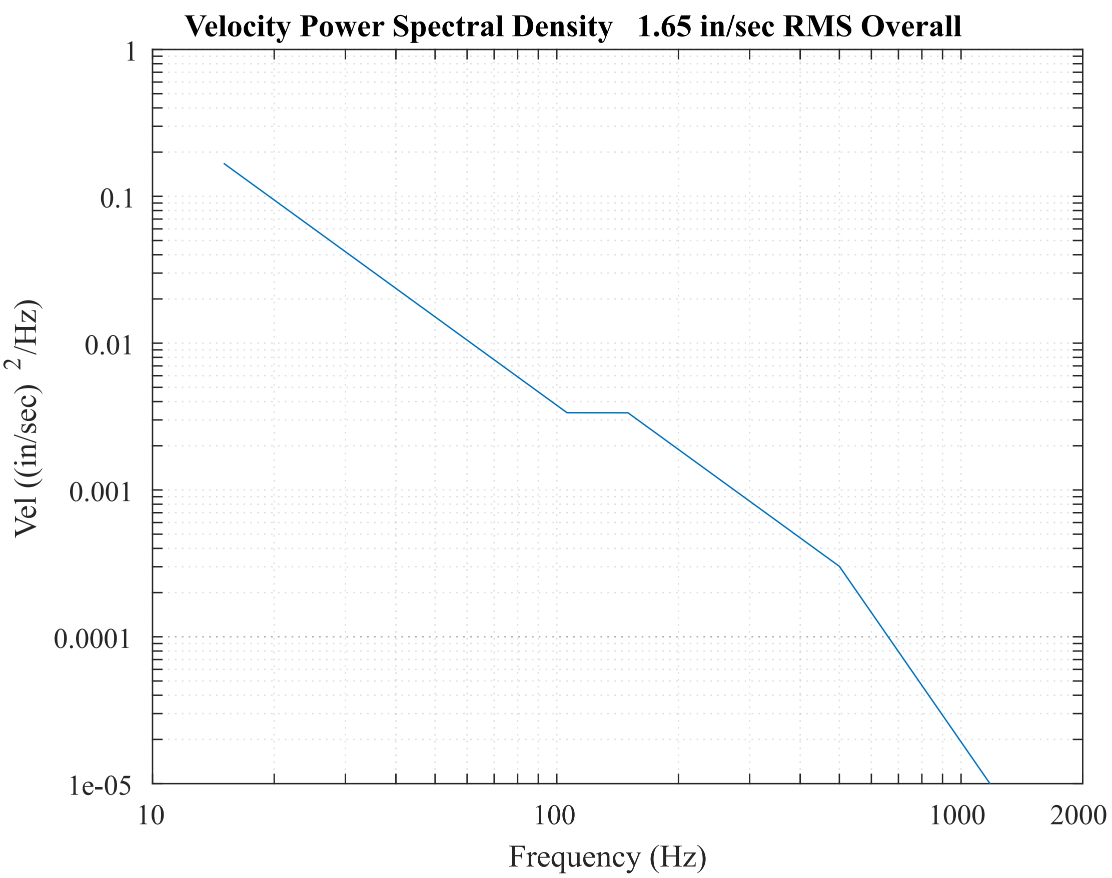
Another helpful feature of PSDs is how easy it is to then convert an acceleration PSD to a corresponding velocity PSD and a displacement PSD. Let:

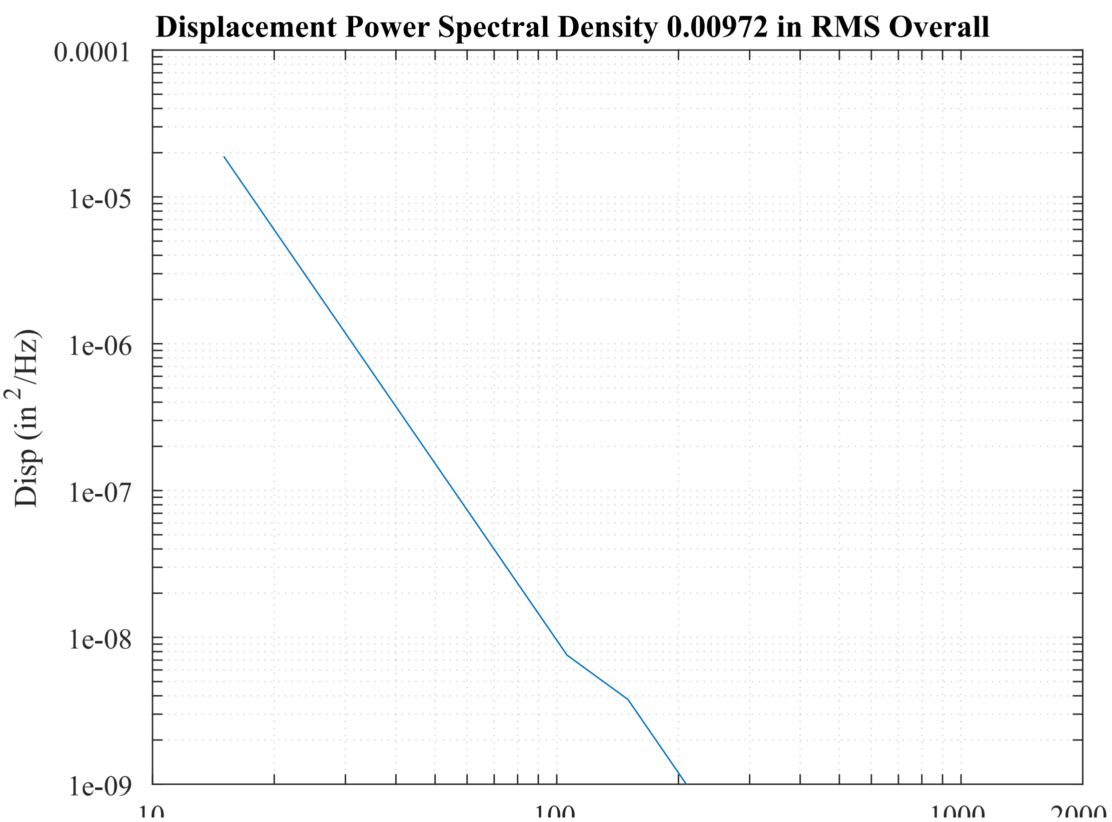
- APSD = Acceleration PSD
- VPSD = Velocity PSD
- DPSD = Displacement PSD

The integration formulas are:

$$VPSD(\omega) = APSD(\omega) / \omega^2$$

displacement PSDs from the jet cargo MIL-STD-810.





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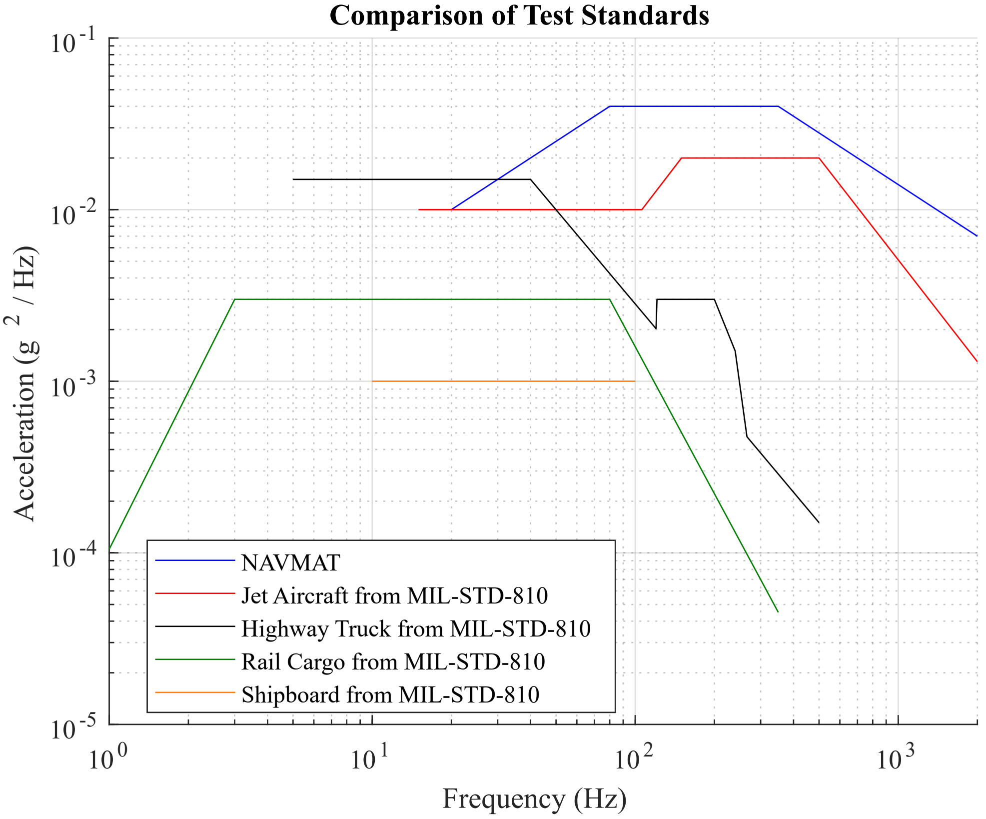
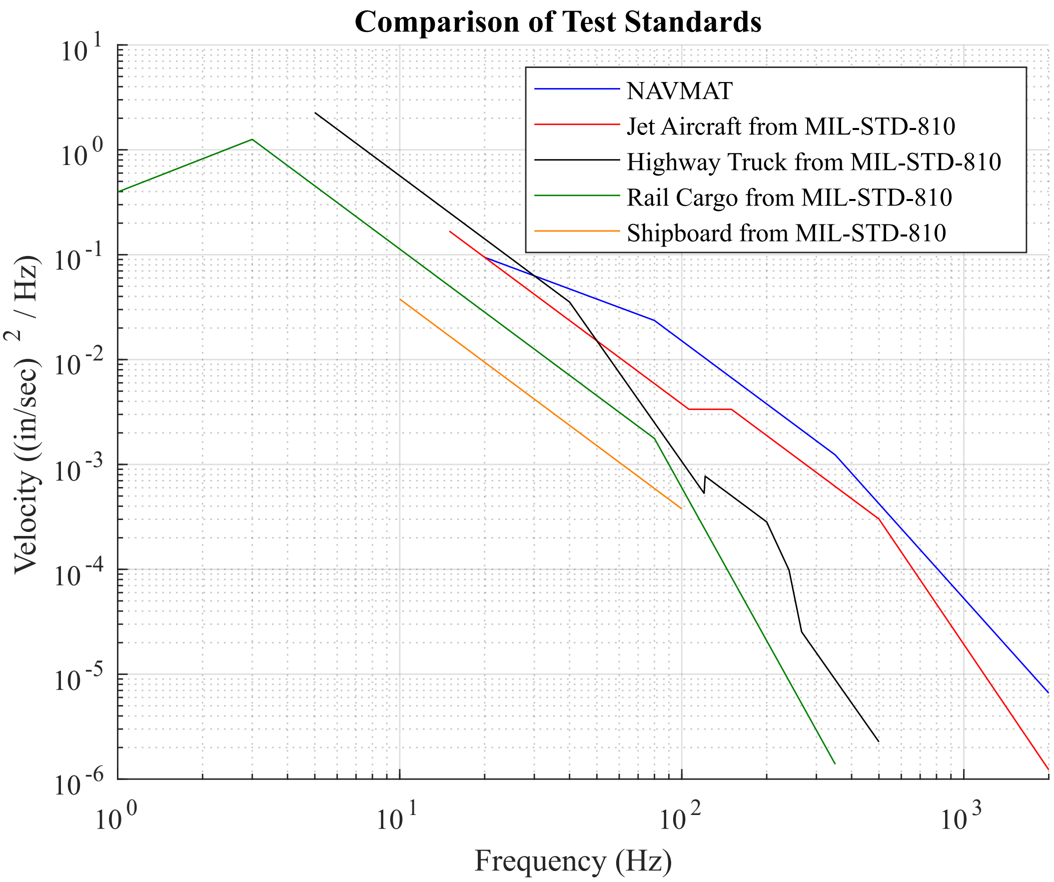

Comparison of Test Standards
Acceleration (g 2 / Hz)
Frequency (Hz)
NAVMAT
Jet Aircraft from MIL-STD-810
Highway Truck from MIL-STD-810
Rail Cargo from MIL-STD-810
Shipboard from MIL-STD-810


Comparison of Test Standards
Velocity ((in/sec) 2 / Hz)
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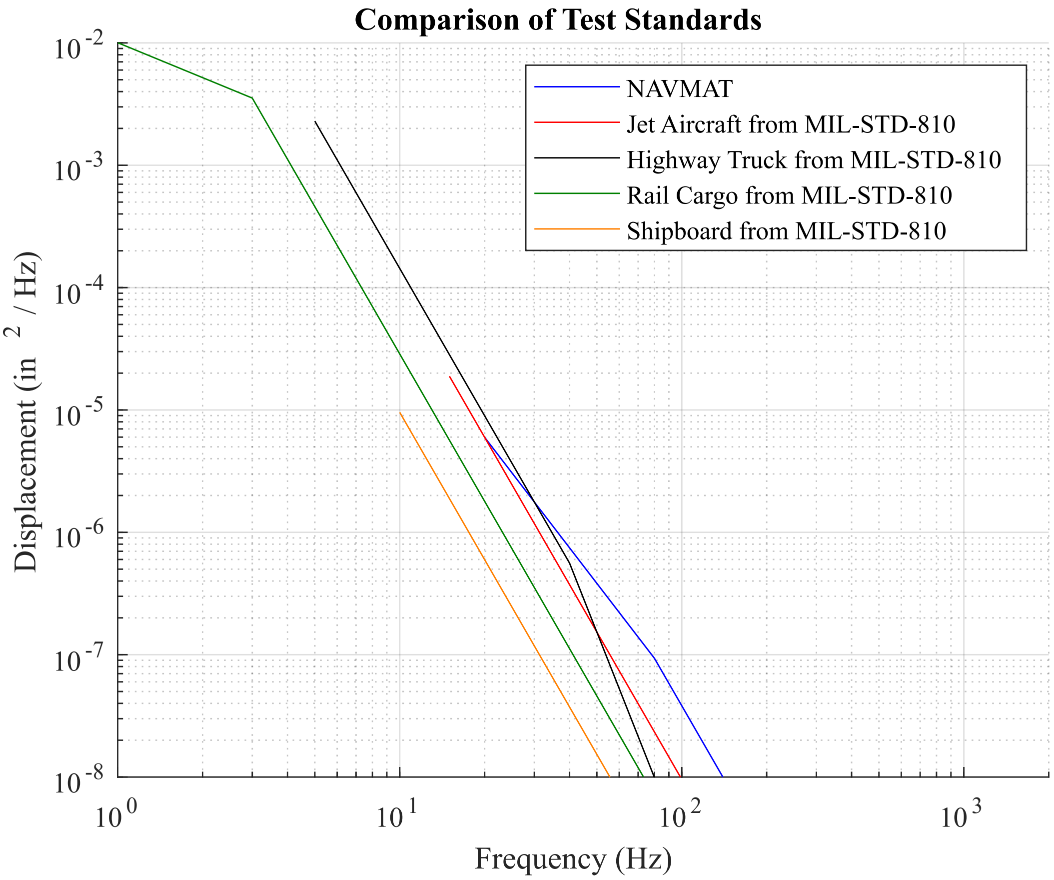
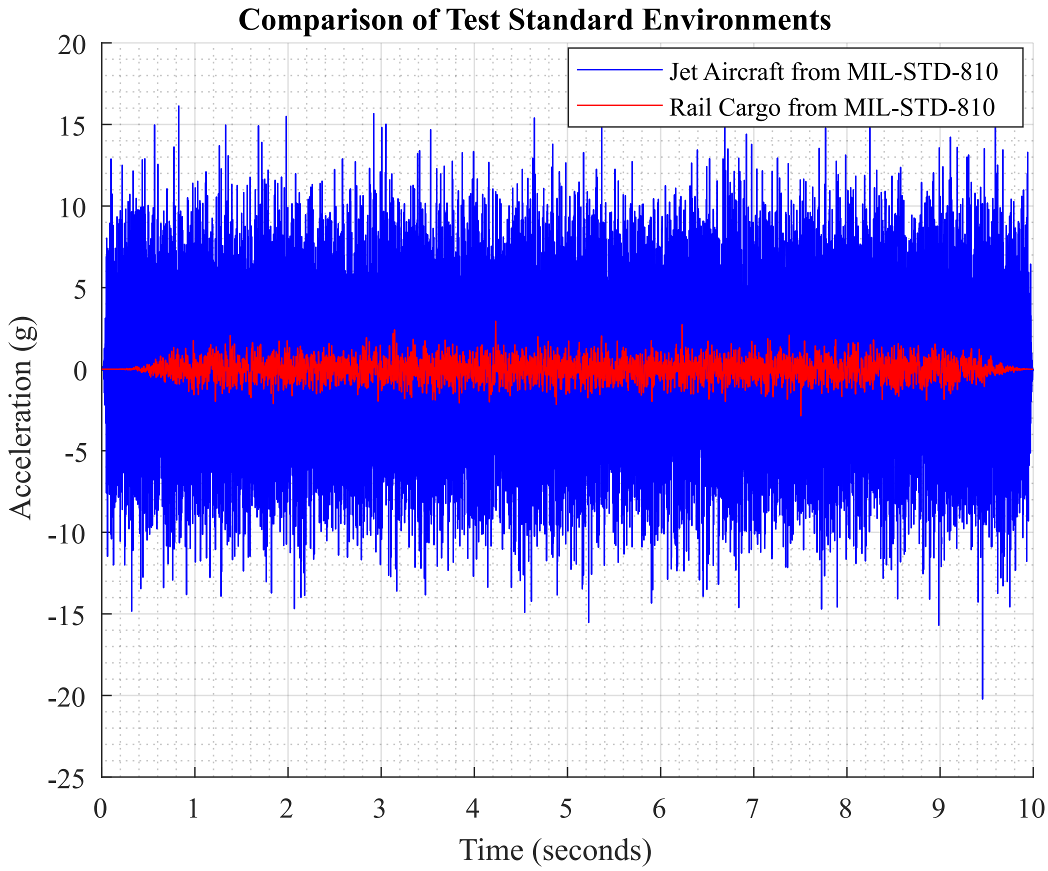
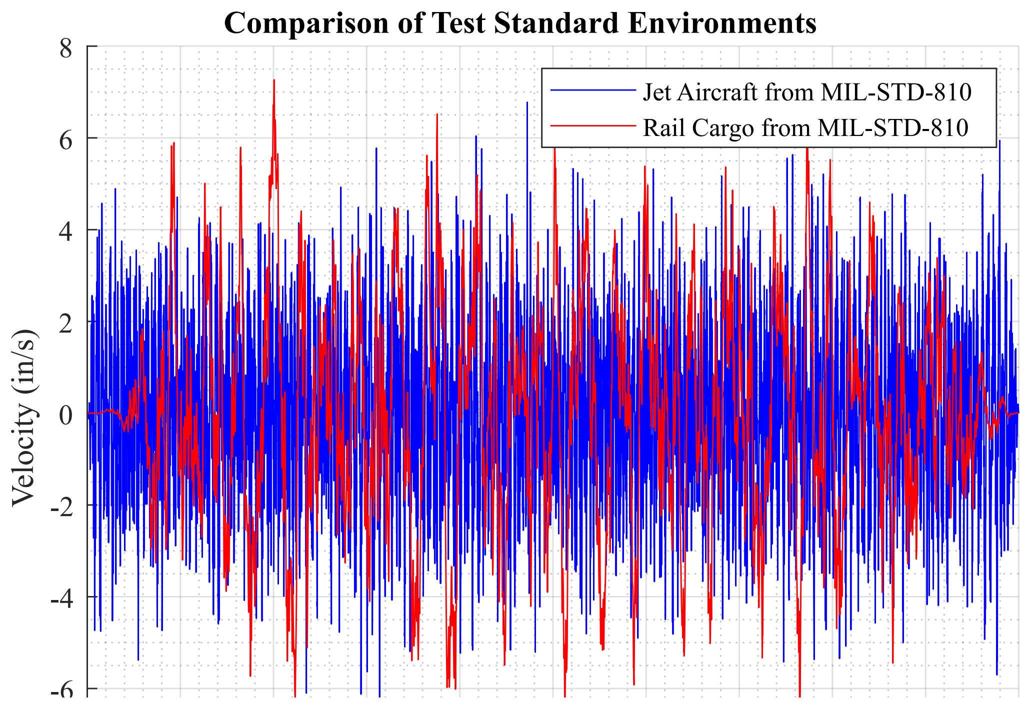


Download the Free VibrationData Toolbox GUI here.

This highlights an obvious benefit of using PSDs... not all frequencies can be treated equally! The vibration environment for rail cargo looks less severe than the jet environment. But by recognizing the lower frequency density of the rail environment, you can predict it will introduce higher velocities and displacements than the jet environment. Let's bring back the time domain of both the jet and rail environment:



Here one would say that the jet aircraft has a more severe environment, but when we look at the velocity environment they have a similar amplitude. And kinetic energy is proportional to velocity squared, so we should arguably be more concerned with this when worried about fatigue.



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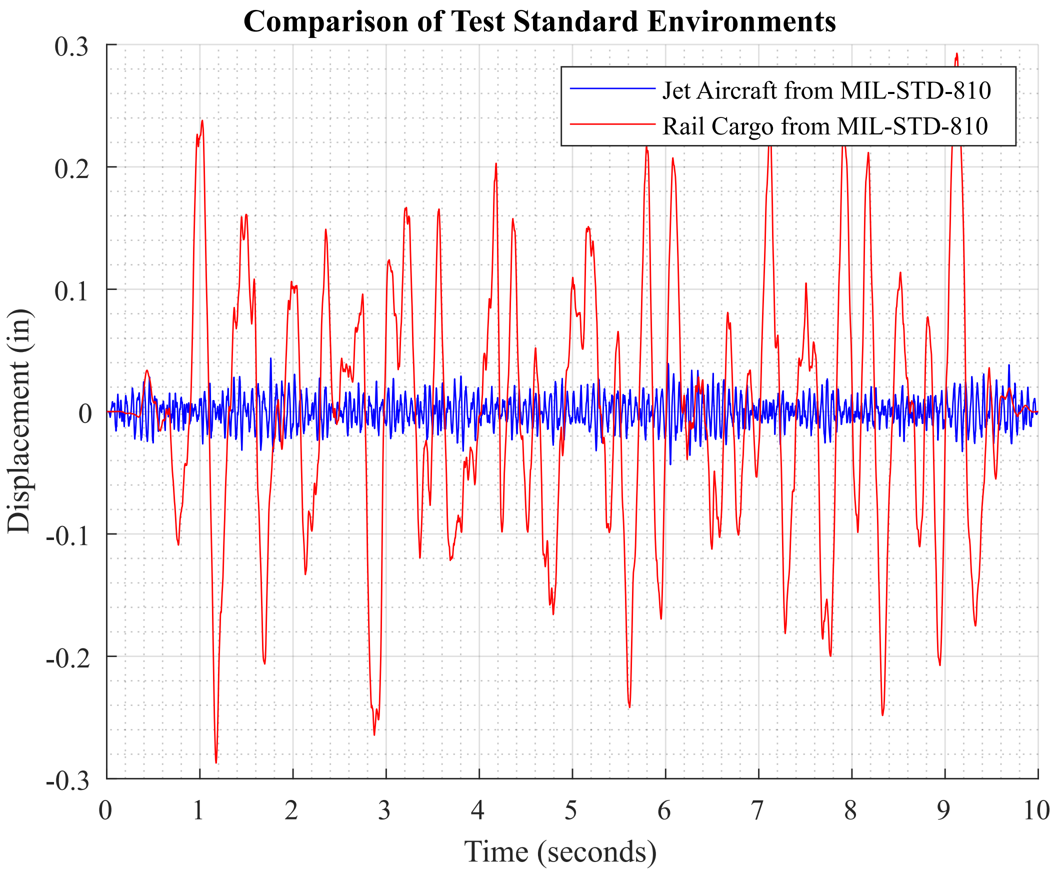
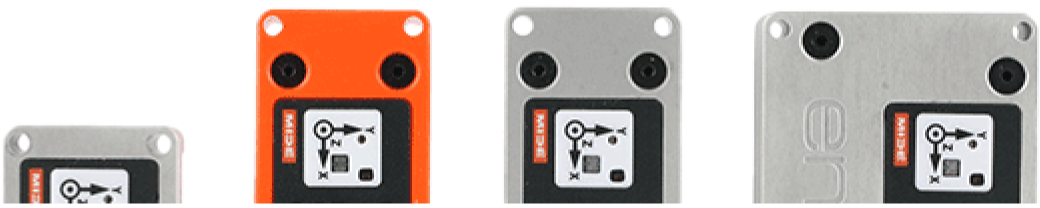
And now when we convert all the way to displacement we can see just how much "stronger" that rail environment is!



Tools to Generate a PSD

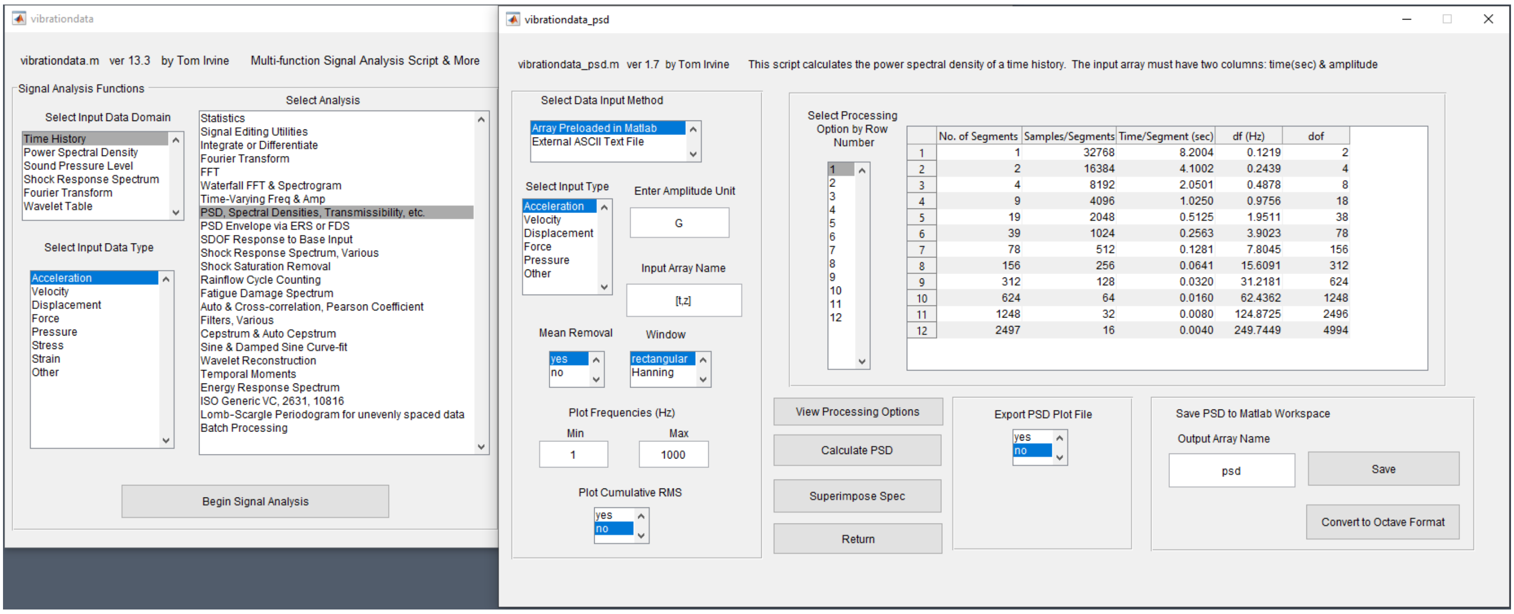
Capture Vibration Data

You'll need some acceleration time series data to use to calculate your own PSD. In order to get that data you can use some traditional accelerometers (see our article on accelerometer types and where to buy them) or you could consider a vibration data logger (see our article on the top 11). Pictured here are our portable and multifunctional enDAQ sensors that were designed to be brought out into the field to quantify vibration environments!



Calculate the PSD with Free Software

All of the plots in this article were generated using our VibrationData Toolbox. You can download this software directly from our website. We also have a dedicated VibrationData Toolbox section in our Help Center with user guides to help you use this software to its full potential. Pictured here is the dialog for calculating PSDs off time series vibration data where it also enables you to set the desired frequency bin width.



Conclusion & Additional Resources

I hope this article helped you better understand how to effectively utilize PSDs to quantify a vibration environment and that you now have the tools to calculate one for your own vibration testing. Here's a quick-hitting summary of the key points!

- **PSDs are preferred to FFTs for vibration analysis** because they are normalized to their bin width allowing you to compare PSDs between environments, test instrumentation, settings and duration and ensure you are comparing apples-to-apples!
- Once you have the PSD there are a lot of different follow-on analysis steps that can be done relatively easily such as:
    - Converting to velocity or displacement
    - Calculating vibration response spectrums to inform your design decisions
    - Reproducing the environment on a shaker system

For more information on how to develop clean PSDs, check out my blog on How to Develop a Vibratior

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Steve Hanly

Steve is the Vice President of Product at Mide. He started out at Mide as a Mechanical Engineer in 2010. He enjoys getting his hands dirty to do some...

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COMMENTS (14)

Ridhwan Nik *5/10/2020, 10:22:04 AM*

Excellent explanation about vibration analysis and PSD.

Reply to *Ridhwan Nik*

Steve Hanly *5/11/2020, 9:49:57 PM*

Thank you Ridhwan, I'm glad you found it helpful!

Reply to *Steve Hanly*

Victor van Spaandonk *5/20/2020, 5:23:41 PM*

Thank you for your explanation!

One thing I don't understand yet is your claim that FFT's can't compensate their amplitude to the duration of the time series data set.
We can normalize the result of an FFT by the length of the time series data set right? This allows to compare the FFT magnitudes for specific frequencies of differently sized data sets.

An FFT is still is not scaled by the size of the frequency bin width indeed, while the PSD is. So the noise contained within 1 bin will be larger for bigger frequency bins (i.e. smaller data sets). So this would have an indirect effect on the amplitude.

Hi Victor,

I just added a section in the blog comparing the FFT to the PSD in a fictitious waveform I generated that had sine tones on noise. Yes the FFT will exacerbate the noise level changing between time lengths - but it also distorts even the very pure sine wave peak's amplitude slightly.

Normalizing the FFT to the length of the time series is a good idea but you still may have issues if you're comparing another dataset that was sampled at a different rate. And normalzing the FFT to the length of the time series is already a step toward a PSD - I'd just recommend to take that final step and normalize to the bin width and make it a PSD.

Hope this helps!

Reply to *Steve Hanly*

Yanuar Sulaiman *11/23/2020, 12:29:27 PM*

I am very interesting with vibration tools Box and just download event I haven't Matlab. I ll traying to use tools box to analysisi earthquake signal to know the background noise of sensoe (Seis mometer). Thank You

Reply to *Yanuar Sulaiman*

Aisha Javed *12/10/2020, 12:16:24 PM*

Hello, thank you for your amazing blogs. I have some questions regarding vibration analysis using video imagery. Actually, I detected and tracked the vibration of a cylindrical structure using videos but the problem is that my final data is in a pixel unit and I need to convert it to a displacement unit. I have searched a lot regarding this but I didn't found much information that how to convert from pixel unit to displacement unit. Secondly, if i use the pixel unit data and time, will it affect my frequency analysis results? Because I am not good at frequency analysis. It's my first time in this area.
A suggestion from you will be really helpful.

Thank you

Reply to *Aisha Javed*

Steve Hanly *12/11/2020, 7:37:49 AM*

Here's one example, they may know more but they may charge you!
http://www.betamachinery.com/services/motion-amplification-vibration-analysis

My recommendation would be to try and find something on the screen that is a fixed length (not vibrating) and then how many pixels make up this length. Then go measure the object and you should be good! You'll want this object to be roughly the same distance away from the camera as the vibrating item you're trying to quantify.

Good luck!

Reply to *Steve Hanly*

Jamilur Reza *2/3/2021, 4:24:56 PM*

Your article is excellent. Can you give me a reference of book or paper to just link up?

Reply to *Jamilur Reza*

Steve Hanly *2/3/2021, 9:16:42 PM*

Awesome, I'm glad you liked it!

I'd recommend Tom's book (free) as a good resource: https://endaq.com/pages/abstract-introduction-to-shock-and-vibration-response-spectra

There is also this book on Random Vibration that has some good stuff:
https://onlinelibrary.wiley.com/doi/book/10.1002/9781118931158

Reply to *Steve Hanly*

NU`MAN AHMAD FAWZAL *3/8/2021, 2:51:16 PM*

Hi Steve,
Thanks for sharing this post which I found very interesting and benefit.
May I know, what is the best way if I would like to compare a vibraiton (FFT format) that i've measured in rotating equipment application with 3 directions (H,V,A) in one format. I mean, all the 3 FFT results combine into single display chart.

Is the PSD format can apply on above case?

Reply to *NU`MAN AHMAD FAWZAL*

You can calculate a resultant by doing the root sum of squares. This can be done for either a FFT or PSD.

Reply to *Steve Hanly*

Nu'man Ahmad Fawzal *3/10/2021, 3:34:51 PM*

Thank you Steve for your reply.
Now it getting more intresting. I've downloded the software that you've shared (Vibration Toolbar). Really workable.

But to get the idea of your advise, could you please share with me the setp on how to get it using your software. If you dont mind, I can share with you my measurement data on rotating equipment with certain defect. The measurement was made in Acceleration (timewaveform).

Can I get you private email?

Reply to *Nu'man Ahmad Fawzal*

Steve Hanly *3/10/2021, 7:30:31 PM*

Nu'man,

I can't provide one-on-one assistance with folks that aren't customers of our products or services but I think I can still help!

I recorded a video of using the Toolbox and doing what I think you need, here it is:
https://www.youtube.com/watch?v=Ro43yMt-yy0

There are more videos online too of various features in the toolbox that should help:

https://endaq.com/pages/shock-and-vibration-analysis-video-tutorials

Thanks,
Steve

Reply to *Steve Hanly*

Nu'man Fawzal *3/15/2021, 9:55:48 AM*

understand the usage of the software and the difference between of all the output selection.

Thank you for sharing this apps.

Reply to *Nu'man Fawzal*